

Quiz 3

● Graded

Student

Samyak Sanghvi

Total Points

3.5 / 4 pts

Question 1

Q1

2 / 2 pts

✓ + 0.5 pts Saying false

✓ + 1.5 pts Valid counterexample

+ 0 pts No/Wrong answer

Question 2

Q2

1.5 / 2 pts

Convolution with box

✓ + 0.5 pts $\text{FT}(\text{box}) = \text{sinc}$

✓ + 0.5 pts multiplication in frequency domain

Result after sampling

✓ + 0.5 pts impulse train

+ 0.5 pts convolution in frequency domain

+ 0 pts No/Wrong answer

Name: Samyak Sanyal

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1. "Applying the same intensity transformation to all three components of an RGB image changes the luminance but preserves the chromaticity of pixels in the image." True or false? Justify your answer.

false

⇒ Let us apply gamma transform on a

tuple (X, Y, Z) s.t. $X \neq Y \neq Z$

we are assured that all 3 numbers will be pushed differently

∴ (x', y', z') will not have chromaticity as (X, Y, Z)

2. A continuous signal $f: \mathbb{R} \rightarrow \mathbb{R}$ is convolved with a box h of unit width and height, and the result $g = f * h$ is sampled at unit spacing to give $\tilde{g}$. What is its Fourier transform $\tilde{G} = \mathcal{F}(\tilde{g})$, in terms of $F = \mathcal{F}(f)$?

$$h = \begin{array}{c} \text{1} \\ \text{---} \end{array}$$

$$h = \begin{cases} 1 & |x| \leq 1/2 \\ 0 & \text{else} \end{cases}$$

$$H = 1 \cdot \text{sinc}(\pi u)$$

$$\tilde{g} = g * s_1$$

$$\therefore \tilde{G} = F \cdot H \cdot S_1$$

$$\tilde{G}(u) = F \cdot \text{sinc}(\pi u) \cdot S_1$$

$$= \begin{cases} F(0) & \text{if } u=0 \\ 0 & \text{else} \end{cases}$$